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A Causal Evaluation of Timeouts in the NBA

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Abstract

There is a long-standing belief in the NBA that calling a timeout during an opponent's scoring run can disrupt their momentum and improve the calling team's chances of winning. However, the empirical evidence in support of this belief is limited and inconclusive. In this paper, we use play-by-play data from the 2021-2026 NBA seasons to conduct a causal analysis of the effect of timeouts on team performance.

Keywords: Causal inference, sports analytics, NBA, timeouts, momentum, difference-in-differences, synthetic control

1 Introduction

Here is a citation **?**.

Acknowledgments and Disclosure of Funding

All acknowledgements go at the end of the paper before appendices and references. Moreover, you are required to declare funding (financial activities supporting the submitted work) and competing interests (related financial activities outside the submitted work). More information about this disclosure can be found on the JMLR website.

Appendix A.

In this appendix we prove the following theorem from Section 6.2:
Theorem *Let u, v, w be discrete variables such that v, w do not co-occur with u (i.e., $u \neq 0 \Rightarrow v = w = 0$ in a given dataset $\mathcal{D}$). Let N_{v0}, N_{w0} be the number of data points for which $v = 0, w = 0$ respectively, and let I_{uv}, I_{uw} be the respective empirical mutual information values based on the sample $\mathcal{D}$. Then*

$$N_{v0} \;>\; N_{w0} \;\;\Rightarrow\;\; I_{uv} \;\leq\; I_{uw}$$

with equality only if u is identically 0. ■

Appendix B.

Proof. We use the notation:

$$P_v(i) \;=\; \frac{N_v^i}{N}, \quad i \neq 0; \quad P_{v0} \;\equiv\; P_v(0) \;=\; 1 - \sum_{i \neq 0} P_v(i).$$

These values represent the (empirical) probabilities of v taking value $i \neq 0$ and 0 respectively. Entropies will be denoted by H. We aim to show that $\frac{\partial I_{uv}}{\partial P_{v0}} < 0$....

Remainder omitted in this sample. See http://www.jmlr.org/papers/ for full paper.

References